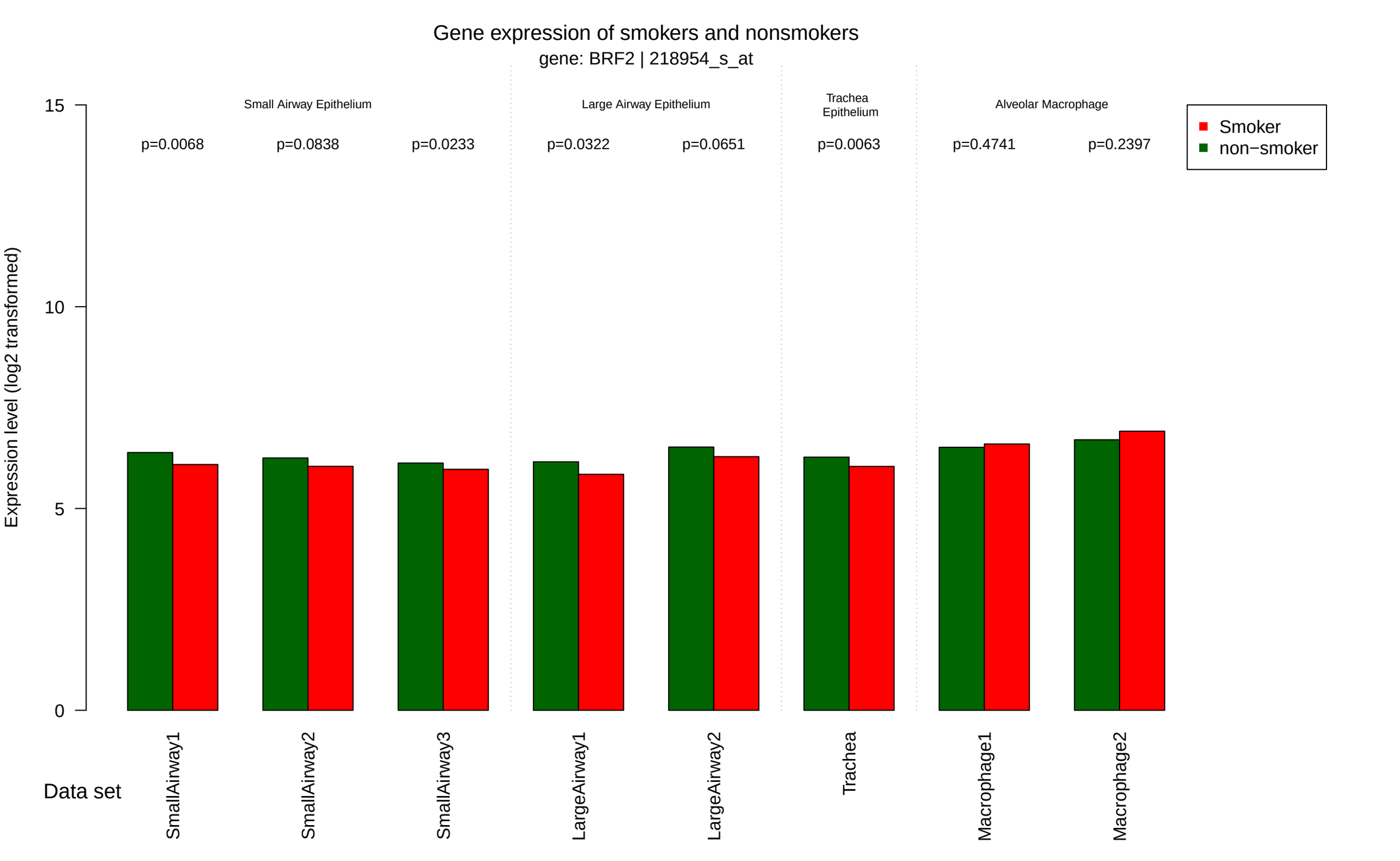
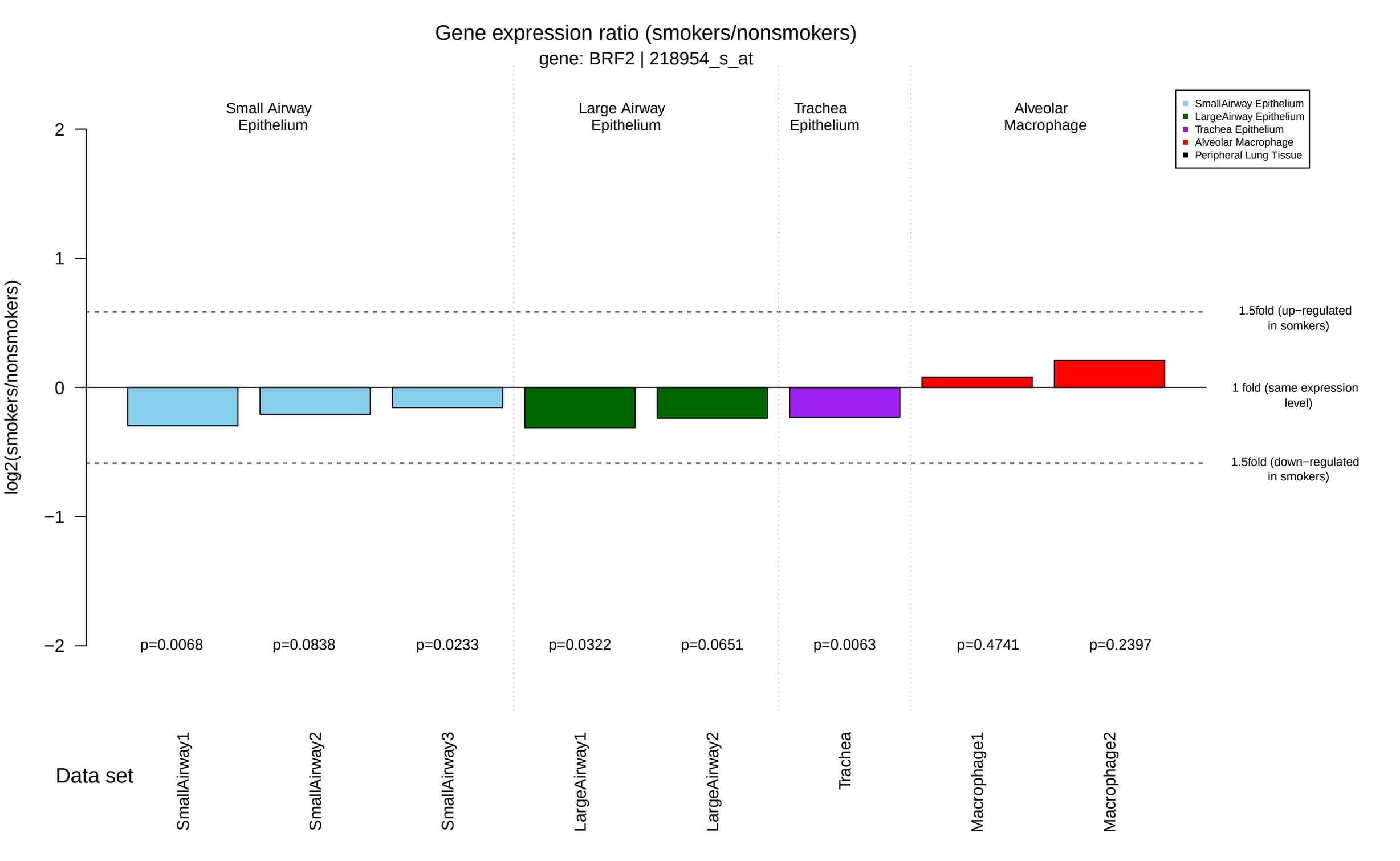
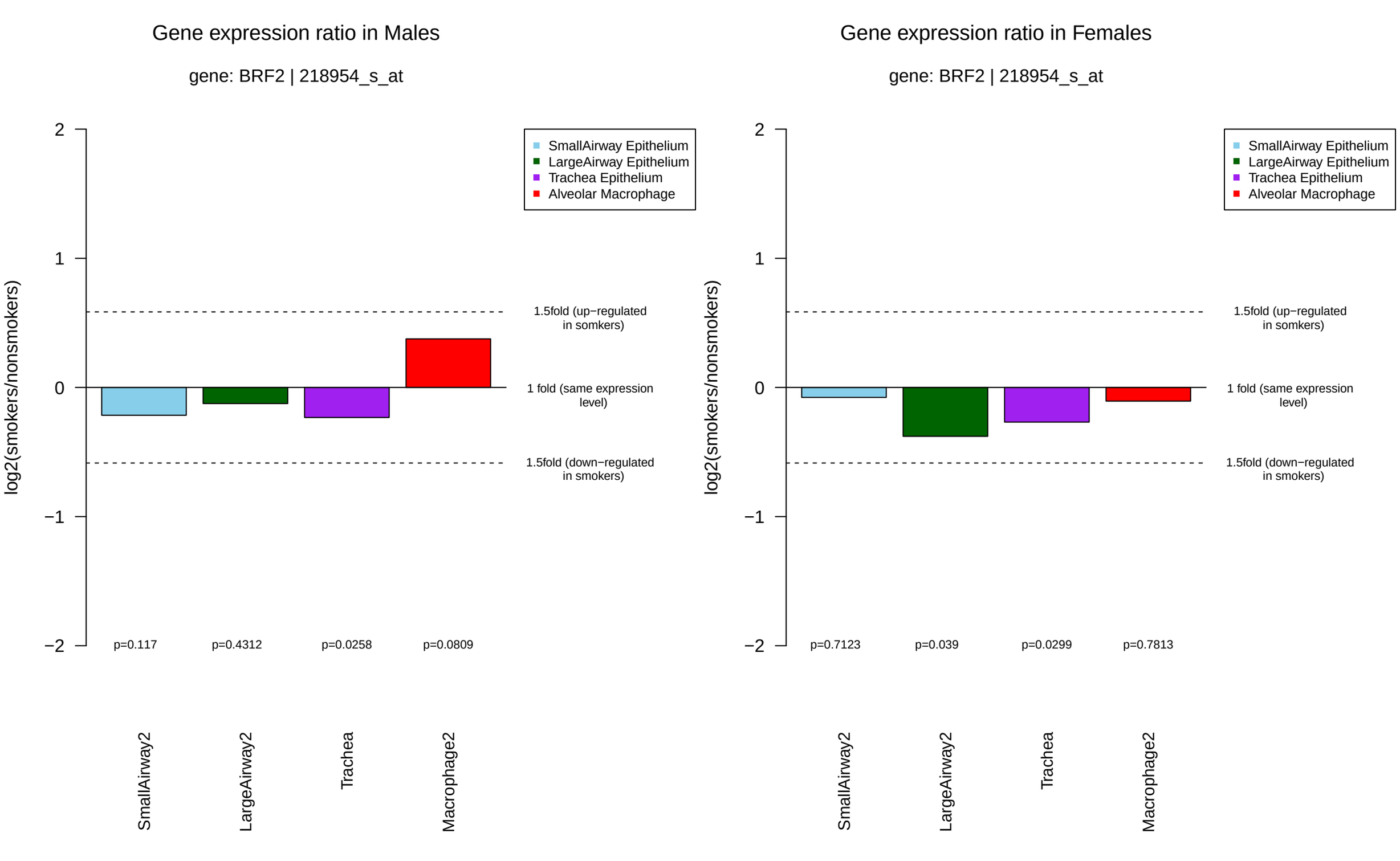
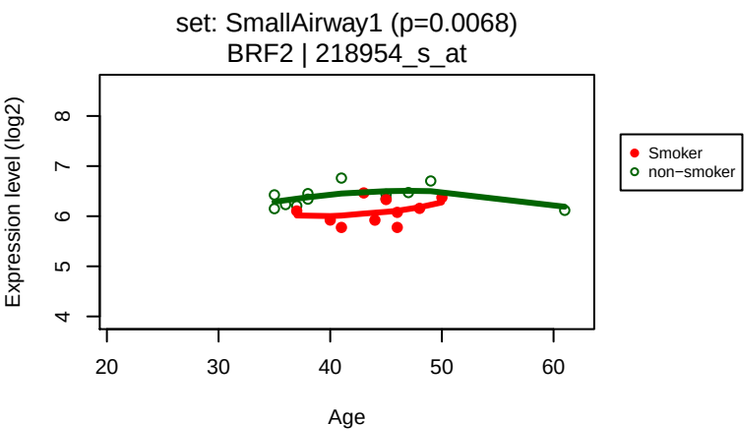
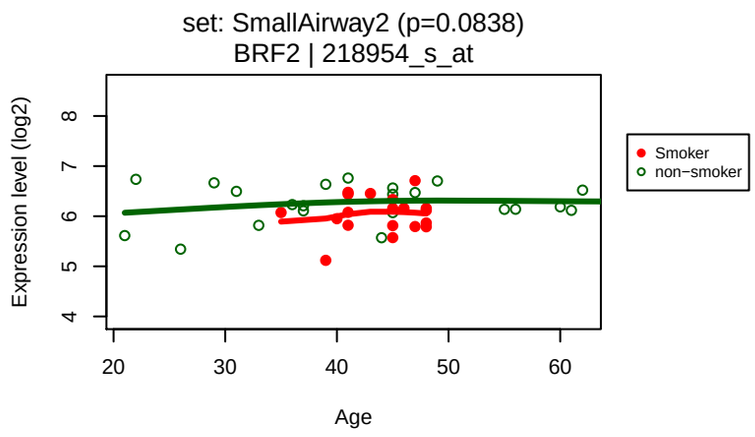
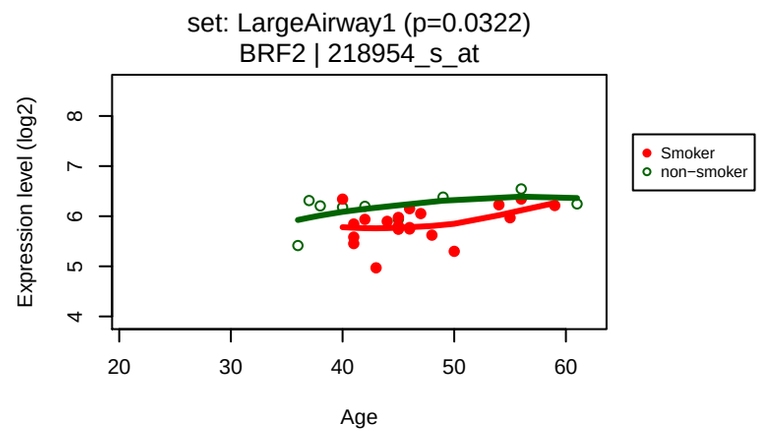

Gene expression of smokers and nonsmokers
gene: BRF2 | 218954_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0068
p=0.0838
p=0.0233
p=0.0322
p=0.0651
p=0.0063
p=0.4741
p=0.2397
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

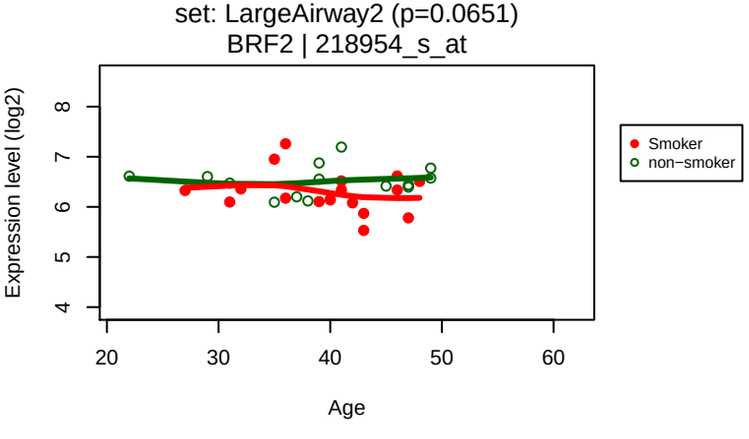

Gene expression ratio (smokers/nonsmokers)
gene: BRF2 | 218954_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0068
p=0.0838
p=0.0233
p=0.0322
p=0.0651
p=0.0063
p=0.4741
p=0.2397
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

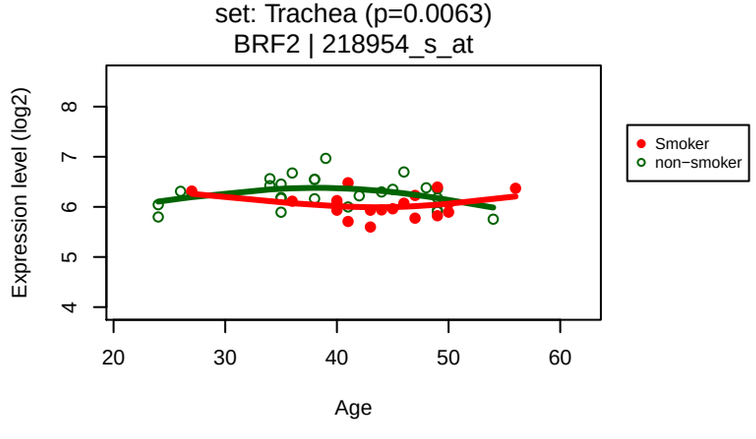

Gene expression ratio in Males
gene: BRF2 | 218954_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.117
p=0.4312
p=0.0258
p=0.0809
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRF2 | 218954_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7123
p=0.039
p=0.0299
p=0.7813
SmallAirway2
LargeAirway2
Trachea
Macrophage2

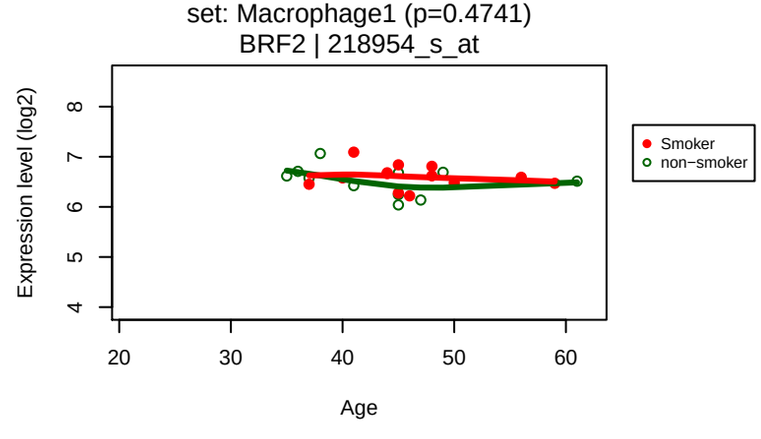

set: SmallAirway1 (p=0.0068)
BRF2 | 218954_s_at
Expression level (log2)
Age
Smoker
non−smoker

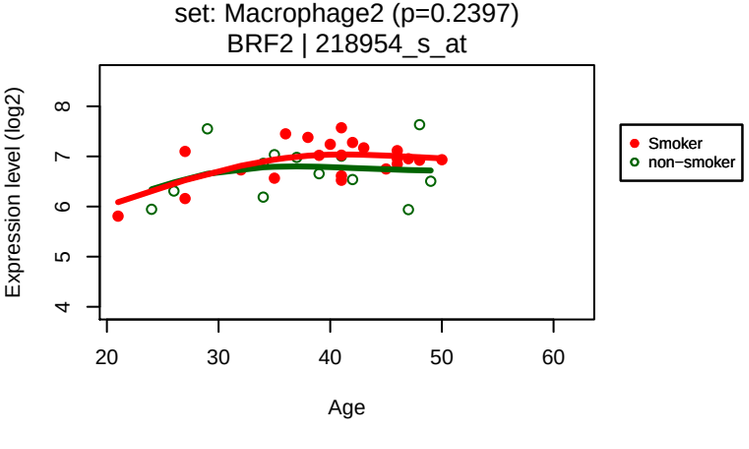

set: SmallAirway2 (p=0.0838)
BRF2 | 218954_s_at
Expression level (log2)
Age
Smoker
non−smoker

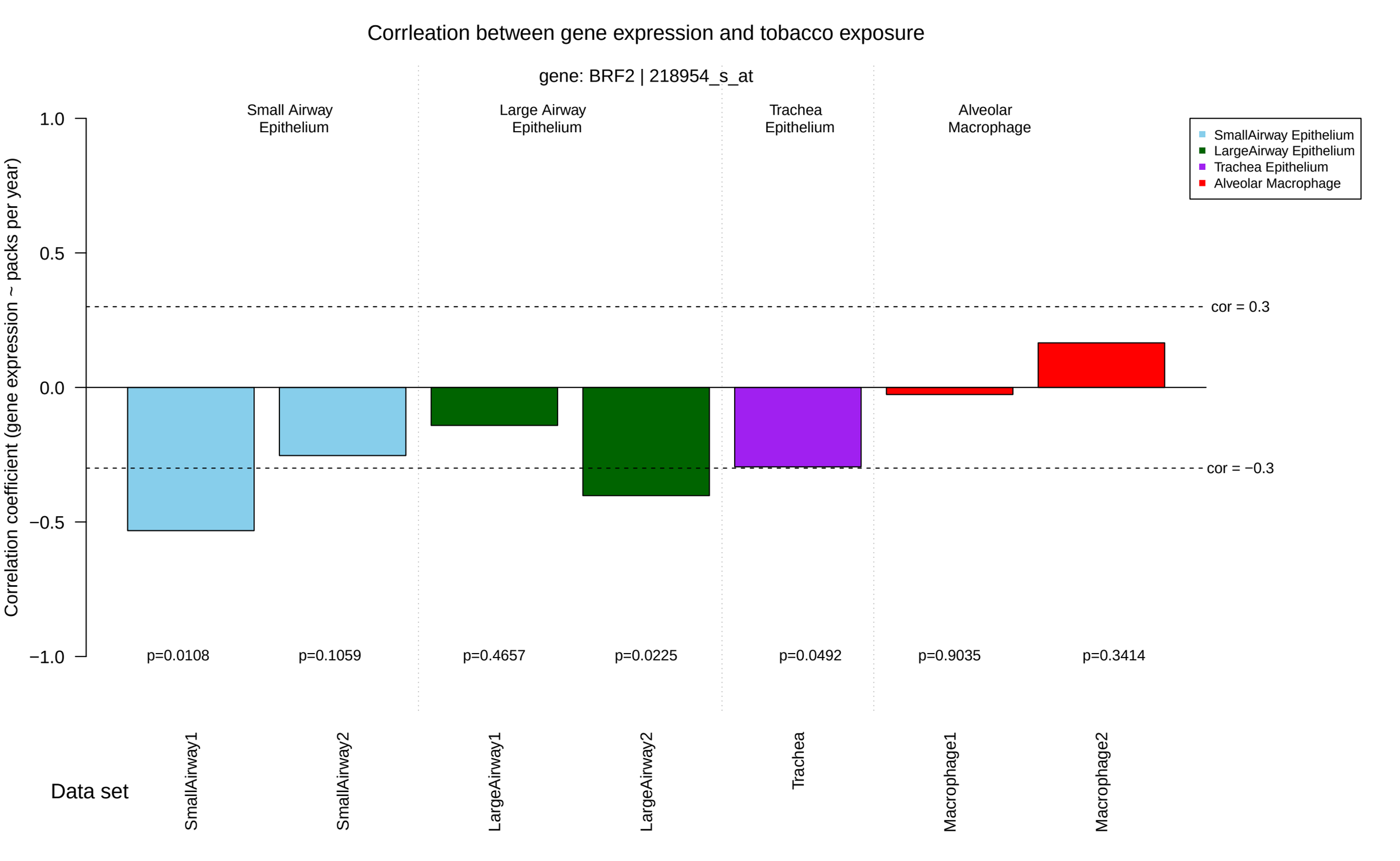

set: LargeAirway1 (p=0.0322)
BRF2 | 218954_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0651)
BRF2 | 218954_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0063)
BRF2 | 218954_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4741)
BRF2 | 218954_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2397)
BRF2 | 218954_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRF2 | 218954_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0108
p=0.1059
p=0.4657
p=0.0225
p=0.0492
p=0.9035
p=0.3414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2